IDEAS 4.0 • PROBLEM STATEMENT 6: AI FOR SMARTER HEALTHCARE

PhysioPulse AI

Upload Doctor Reports → AI Auto-Plans Sessions → 60 FPS On-Device 3D Vision Biofeedback

Patients upload discharge summaries, MRI findings, or prescription sheets. PhysioPulse simplifies the clinical jargon, formulates personalized kinetic rehab sessions, and coaches form in real time using any laptop or phone camera.

33-Point 3D Pose

60 FPS LOCAL CORE

MediaPipe Full Landmarker running purely on-device. Zero cloud video latency; 100% patient video privacy.

Report Upload & AI Plan

INTAKE → AUTO-PLAN

Patients upload doctor/MRI reports. AI extracts restrictions and plans auto-advancing kinematic queues.

8 Clinical Protocols

FULL-BODY SUITE

Cervical, Shoulder, Elbow, Wrist, Lumbar, Hip & Knee with live angle & vector goniometry.

100% On-Device Safety

RESPONSIBLE AI

Built-in red-flag emergency triage (VAS ≥8 locks unguided reps) with CMS RTM billing ledger.

THE REAL-WORLD ORTHOPEDIC CRISIS

The Problem: Unintelligible Reports & The 'Blind Home-Rehab' Gap

The ₹1,000/Session Cost Barrier

70%+ REHAB DROP-OFF RATE

- Daily clinic physiotherapy costs ₹800–₹1,500 per visit plus painful travel in Tier-2/3 towns.
- Most patients abandon therapy within 10 days, causing permanent loss of joint Range of Motion (ROM).

Biomechanical Compensation & Cheat-Reps

ZERO EYES AT HOME

- Patients unconsciously cheat when recovering—shrugging shoulders or twisting spines.
- Passive videos (YouTube/fitness apps) cannot see joint angles or interrupt a patient mid-rep when form slips.

Doctor Report Upload & Autonomous Planning

BRIDGING THE PAPER GAP

- Patients leave clinics with unintelligible MRI summaries or paper exercise handouts.
- PhysioPulse lets patients upload doctor discharge reports directly—AI translates clinical jargon and auto-plans personalized kinematic sessions.

THE CORE REALIZATION: Text chatbots only offer Q&A. True 'Smarter Healthcare' demands combining LLM Clinical Triage with 60 FPS On-Device Computer Vision Biomechanics to guide recovery hands-free.

MANDATORY TRACK REQUIREMENTS

100% Alignment with Track 6: 'AI for Smarter Healthcare'

1. Multilingual AI Health Assistant (Hindi, English + Regional Dialect)	Natural language symptom intake supporting Hindi (हिन्दी), Hinglish, and English. Accompanied by voice biofeedback coaching the user during every movement.	**Intake Stage + pyttsx3 Audio Coach**
2. Smart Symptom Guidance & Emergency Medical Triage	Evaluates VAS pain scale (1–10). Built-in red-flag emergency guardrail: acute pain (≥8/10) automatically locks unguided exercise and triggers hospital referral.	**AI Triage Guardrail Engine**
3. Medical Report Simplifier & Autonomous Session Planner	Patients upload doctor discharge summaries, MRI scans, or prescription sheets. Clinical AI parses diagnoses and auto-plans sequential kinematic routines ('prescribed_session.json').	**Doctor Report Upload + AI Engine**
4. Preventive Healthcare & Actionable Wellness Tracking	Preventive posture guard for desk workers (P3 Cervical Neck & P8 Carpal Tunnel Wrist) with degree-by-degree ROM progression and compliance charts.	**physiopulse.py + Web Telemetry**
5. Responsible AI, Privacy & Non-Diagnostic Guardrails	100% local on-device camera inference (zero video frames sent to cloud). Explicit clinical disclaimer: PhysioPulse assists recovery—never replaces clinicians.	**Local OpenCV / MediaPipe Core**

AUTONOMOUS END-TO-END WORKFLOW

Our Approach: 4-Stage Closed-Loop AI & Edge Vision Architecture

STAGE 1

Doctor Report Upload & AI

- Upload discharge / MRI reports
- AI simplifies clinical jargon
- Acute red-flag safety triage
- Auto-plans personalized sets

STAGE 2

3D Edge Vision

- 33-landmark MediaPipe Full core
- True gravity reference vector
- Scale-invariant normalization
- 60 FPS local on-device pipeline

STAGE 3

Biofeedback Engine

- 8 clinical orthopedic protocols
- 6-frame compensation debouncer
- Non-blocking audio voice coach
- Red HUD skeleton visual alert

STAGE 4

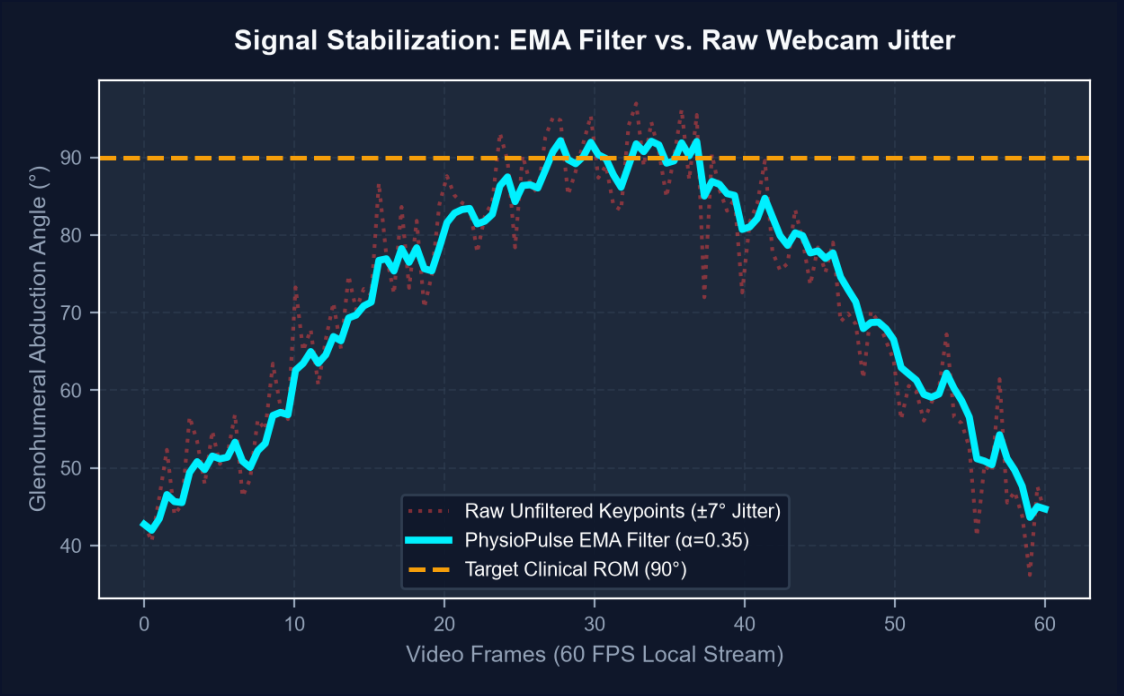
Clinical Telemetry

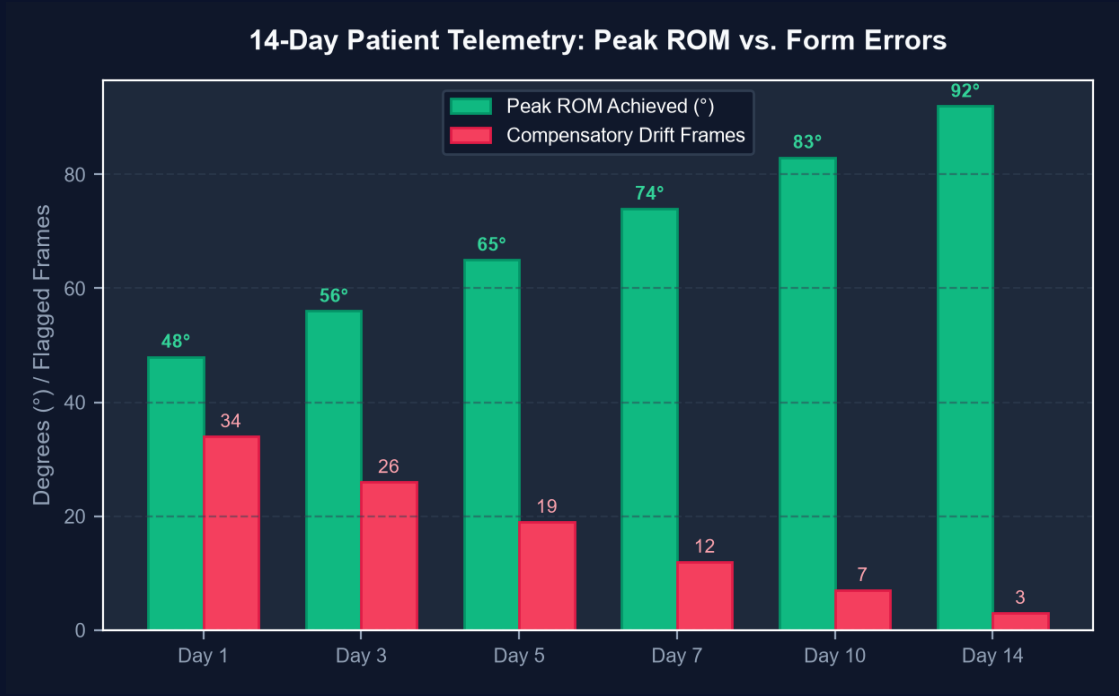
- Auto-advances protocol queue
- Writes clinical_session_report.json
- CMS RTM CPT insurance coding
- 1-Click Doctor PDF letterhead

🔄 **CLOSED-LOOP BIOMECHANICAL ADAPTATION:** Why PhysioPulse is truly closed-loop:

When a patient completes reps with high bilateral symmetry and zero compensation frames, the Clinical AI Agent automatically increases target ROM (+5°) and rep volume for next session. If compensatory shrugs are caught, the system auto-calibrates downward to protect the articular capsule.

Engineering Depth: Signal Stabilization & Clinical ROM Telemetry





2D/3D Dot-Product Goniometry

`θ = arccos((BA · BC) / (||BA|| ||BC||))`

Eliminates arctan quadrant flips; measures true anatomical vertex angles across 3D space.

Dual-Layer EMA Jitter Filter

`α_landmarks=0.38` • `α_angles=0.30`

Filters out webcam pixel noise and tremor without introducing artificial lag or latency.

6-Frame Biomechanical Debouncer

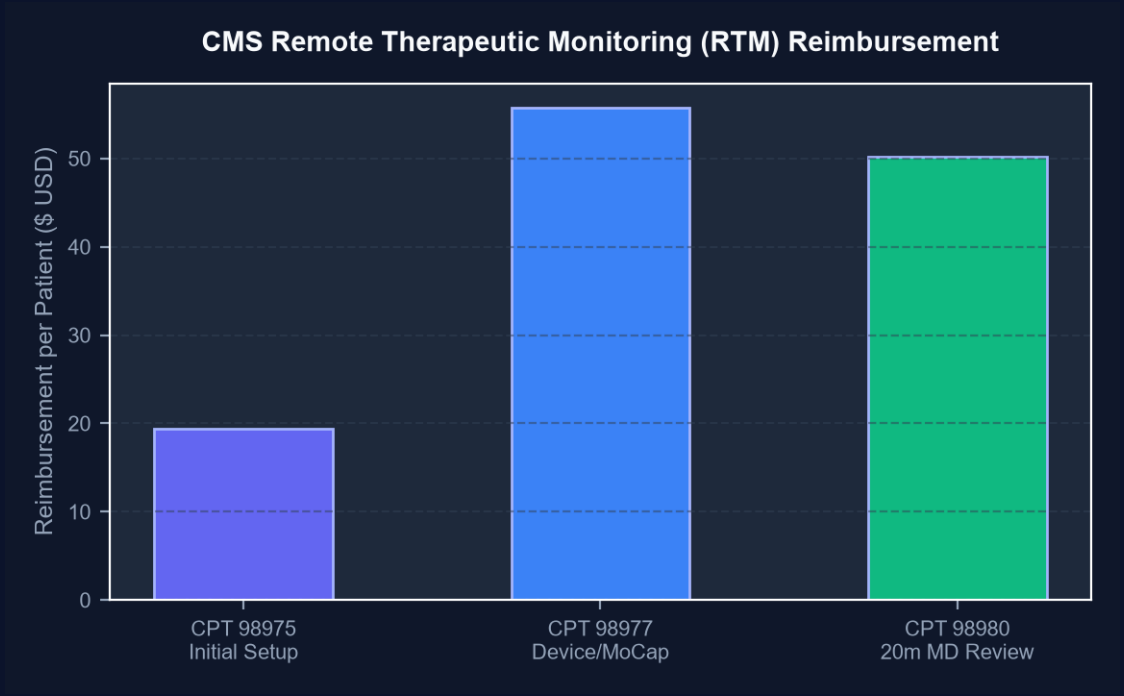
`Duration > 0.2s` • `Confidence > 0.50`

Requires 6 consecutive bad posture frames before alerting, preventing false alarms.

Full-Body Clinical Suite: 8 Parametric Kinematic Protocols

ID	Clinical Protocol Name	Anatomical Joint	Primary Vector & Target	Compensatory Error Caught
P1	**Frozen Shoulder Abduction**	Glenohumeral Joint	Vertical Ref → Shoulder → Elbow (90°+)	**Shrugging / Bent Elbow**
P2	**Post-ACL Knee Extension / Squat**	Tibiofemoral Joint	Hip → Knee → Ankle Flexion (≤112°)	**Torso Forward Lean**
P3	**Cervical & Desk Posture Guard**	Cervical Spine	Collarbone Plumb Line → Nose (<14°)	**Posture Drift / Head Tilt**
P4	**Post-Fracture Elbow Flexion**	Humeroulnar Joint	Shoulder → Elbow → Wrist (145°→55°)	**Upper-Arm Elbow Drift**
P5	**Bilateral Overhead Wall-Slide**	Bilateral Shoulder	Dual Arm Synchronous Abduction (115°+)	**>18° Bilateral Symmetry Gap**
P6	**Total Hip Replacement Abduction**	Coxofemoral Joint	Hip Vertical → Hip → Ankle (28°+)	**Lateral Trunk Lean**
P7	**Lumbar Spine Flexion / Slip-Disc**	Lumbar Spine	Hip Vertical → Hip → Shoulder (25°+)	**Spinal Shoulder Twist**
P8	**Carpal Tunnel Wrist Extension**	Radiocarpal Joint	Elbow → Wrist → Index Knuckle (22°+)	**Shoulder Trapezius Tension**

Commercial Viability: CMS Remote Therapeutic Monitoring (RTM)



Direct Reimbursement under CMS RTM

CPT CODES 98975, 98977 & 98980

PhysioPulse qualifies for official Remote Therapeutic Monitoring reimbursement ($125.28/patient/month). Clinics generate new Medicare/commercial revenue without adding staff.

$150,336 Annual Practice Yield

PREDICTABLE B2B SAAS + CLINIC MARGIN

A moderate clinic cohort of 100 enrolled post-op patients yields $150K+ in annual practice revenue. PhysioPulse charges an 18% SaaS platform fee ($22/pt/mo).

Closing the Doctor Feedback Loop

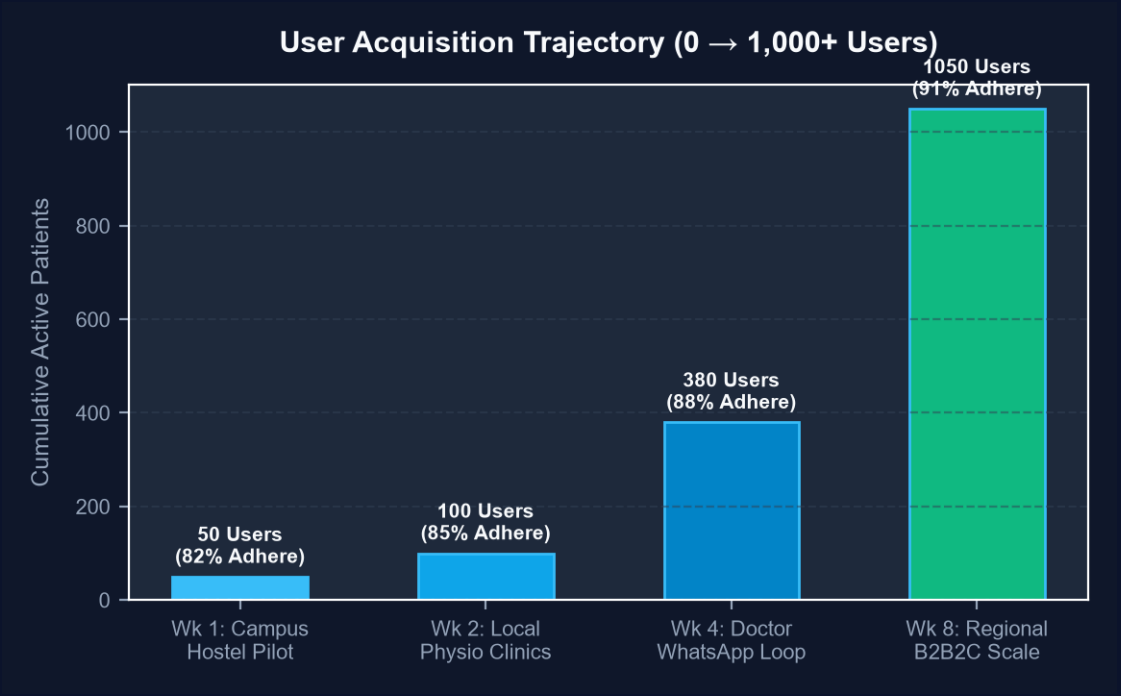
OFFICIAL PDF TELEMETRY LETTERHEAD

Doctors receive automated degree-by-degree ROM progression curves, compliance timestamps, and bilateral symmetry scores—reducing costly malpractice and readmissions.

WHY CLINICS ADOPT PHYSIOPULSE: Traditional hardware systems (wearables/sensors) cost ₹40,000+ per unit and require Bluetooth pairing. PhysioPulse requires zero dedicated hardware—it turns the patient's existing smartphone or laptop webcam into an FDA-compliant remote monitoring portal.

MANDATORY DELIVERABLE #2

Impact, Adoption & Scaling Plan (0 → 1,000+ Users)



Phase 1: First 100 Users (Zero-Budget Pilot)

- Users 1–50 (Campus Hostel Rollout): Deploy P3 (Cervical Posture) & P8 (Carpal Tunnel) across university hostels & coding clubs as a 2-minute daily check.
- Users 51–100 (Local Clinic Partnering): Onboard 5 outpatient clinics in Gurugram/NCR—giving therapists our free link so patients perform supervised home reps.

Phase 2: Scaling to 1,000+ Users (Doctor Viral Loop)

- WhatsApp Telemetry Growth Loop: Every completed session generates a 1-click Doctor Summary shared by the patient on WhatsApp.
- B2B2C Clinic Multiplier: Each orthopedic surgeon or therapist who receives clean reports prescribes PhysioPulse to 25–40 additional outpatients.

HOW WE MEASURE REAL CLINICAL IMPACT (NOT JUST CLICKS):

1. Clinical ROM Gain: Mean joint degree improvement over 14 days (e.g. Frozen Shoulder 48° → 92° validated via JSON telemetry)
2. Form Error Reduction: >65% drop in flagged compensatory drift frames within 7 sessions
3. Adherence Elevation: Lifting home rehab adherence from 30% clinic baseline to >85% via real-time audio coaching

Responsible AI Guardrails, Privacy & Rehearsed 90s Live Demo

STRICT COMPLIANCE WITH TRACK 6 RULES

- **Assists—Never Replaces—Doctors:** All outputs carry explicit disclaimers: PhysioPulse AI is an educational & biomechanical feedback assistant, never a definitive diagnostic authority.
- **100% Local On-Device Video Privacy:** All 33-point pose landmarking runs in local RAM on device. Zero raw patient camera frames ever leave the computer or enter the cloud.
- **Emergency Red-Flag Safety Lock:** If a patient indicates acute pain (VAS ≥ 8/10) or trauma history, the system locks unguided exercise generation and shows emergency hospital contact info.
- **Synthetic Patient Data Guardrails:** All demo profiles, MRI scans, and telemetry progression trajectories use 100% anonymized, synthetic medical records.
- **Hands-Free & Non-Technical Accessibility:** One-click preset cards and natural voice prompts enable elderly patients to complete their routine without touching the mouse or keyboard.

REHEARSED 90-SECOND STAGE DEMONSTRATION

Step 1: Doctor Report Upload & AI Auto-Plan (15s)

Upload hospital discharge summary / MRI report (or click sample report) → AI simplifies medical jargon, extracts restrictions, and auto-plans the tailored session queue.

Step 2: Auto-Synthesize Kinetic Queue (15s)

Click 'Synthesize Prescription Queue' → Clinical AI Agent maps symptoms to P1, P5, P3, P4 with target ROM & rep counts in prescribed_session.json.

Step 3: Launch 60 FPS Local Vision Engine (20s)

Click 'Launch Vision Session' → Camera opens, announces Protocol 1 via voice coach, and displays live shoulder abduction angle in green.

Step 4: Trigger Live Compensatory Error (20s)

Intentionally shrug shoulder or tilt neck → Skeletal vector turns RED instantly on screen while Voice Coach commands: 'Don't shrug your shoulder!'

Step 5: Auto-Advance & Doctor PDF Report (20s)

Complete target reps → Engine auto-advances to next protocol, writes clinical_session_report.json, and renders official printable Doctor Letterhead with CPT codes.